Product Requirements Document

Punarvasan Tracker — Real-Time Slum Redevelopment Progress App

Version: 1.0 **Status:** Draft for Pilot Proposal **Prepared for:** Ward-level pilot (proposed by MLA Varun Sardesai, Bandra East) — scalable to SRA/MHADA/DRP city-wide

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1. Problem Statement

Slum redevelopment in Mumbai (SRA, MHADA, Dharavi Redevelopment Project) routinely stretches across years or decades. Displaced families living in transit camps have **no visibility** into whether their building is actually under construction, delayed, litigated, or abandoned. This creates:

- Distrust between residents, contractors, and government
- No documentary trail when projects stall
- No easy way for elected representatives to answer constituents with data
- No systemic way to compare contractor performance across projects

Insight driving this product: Food delivery apps solved exactly this trust problem for a ₹200 order using live, stage-based tracking. The same pattern applies to a ₹50 lakh home.

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2. Goals

Goal	Metric
Give residents visibility into their building's progress	% of registered families actively checking status monthly
Give government a live, aggregate view across projects	# of projects onboarded to dashboard
Create an accountability trail for contractors	% of projects with on-time milestone updates
Reduce distress/complaints caused by information gaps	Reduction in ward-office visits/RTI queries about project status

Non-goals (for v1): This app does not handle legal/dispute resolution, payments, or eligibility certification — it only tracks and displays construction progress against already-

sanctioned projects.

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3. User Personas

3.1 Resident (Primary user — mobile)

One registered member per eligible family. Low-to-medium digital literacy. Wants to know: *"Is my building being built? When do I get my house?"*

3.2 Contractor (Secondary user — mobile/web)

Registers with an official contract ID tied to a specific SRA/MHADA project. Responsible for updating construction milestones and uploading proof (photos/geotags).

3.3 Government/SRA Official (Tertiary user — web dashboard)

Ward officer, SRA officer, or MLA's office. Needs an aggregate, filterable, exportable view across all projects in their jurisdiction.

3.4 System Admin

Onboards new projects, verifies contractor credentials, manages resident-to-building linkage, handles data corrections.

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4. Scope — MVP (Pilot Phase)

Pilot scope: **2–3 redevelopment projects in one ward**, ~500–1000 registered families, 1–2 contractors.

In scope for MVP

- Resident registration & login (Aadhaar-linked or SRA allotment ID)
- Contractor registration & login (contract ID verification by admin)
- Project → Building → Family hierarchical data model
- Stage-based progress tracker (6 fixed stages, see §6.1)
- Photo/date-stamped milestone updates by contractor
- Resident-facing timeline view (read-only)
- Basic complaint/query submission tied to a unit
- Government dashboard: list + map view of projects, status filter, delay flagging
- Push/SMS notification on stage change

Out of scope for MVP (future phases)

- In-app payments or compensation tracking
- Legal case/litigation status integration
- AI-based photo verification (fraud detection)
- Multi-language voice/IVR interface
- Full analytics/BI reporting suite

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5. User Stories

Resident

- As a resident, I can register using my SRA allotment ID so the app links me to my exact building/unit.
- As a resident, I can see my building's current construction stage and expected next milestone.
- As a resident, I can view dated photos uploaded by the contractor as proof of progress.
- As a resident, I can raise a complaint/query linked to my unit and see its status.
- As a resident, I get a notification when my building's stage changes.

Contractor

- As a contractor, I can register with my contract ID and get linked to my assigned project(s)/building(s).
- As a contractor, I can update the construction stage for a specific building.
- As a contractor, I can upload photos with auto-captured date/geotag as proof.
- As a contractor, I can manage multiple projects from one login via a project switcher.

Government Official

- As an official, I can view all projects in my jurisdiction on one map/list, color-coded by status.
- As an official, I can filter by ward, contractor, scheme type (SRA/MHADA/DRP), or delay status.
- As an official, I can see which contractors are consistently missing milestones across projects.
- As an official, I can export a status report for assembly questions / RTI responses.

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6. Functional Requirements

6.1 Construction Stages (fixed pipeline, mirrors "order tracking" UX)

1. Survey & Eligibility Confirmed
2. Demolition / Site Cleared
3. Foundation Complete
4. Structure — Floor-wise (shows floor number out of total)
5. Plumbing, Electrical & Finishing
6. Possession Ready / Handover

Each stage requires: date marked, contractor note (optional), photo (mandatory from stage 2 onward).

6.2 Data Hierarchy

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Project (SRA/MHADA/DRP scheme)
  └── Building/Wing
        └── Unit/Flat
              └── Registered Family (1 member)
Contractor ── linked to one or more Projects via Contract ID
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6.3 Roles & Permissions

Action	Resident	Contractor	Govt Official	Admin
View own unit progress	✅	—	—	—
View all units in a project	❌	✅ (own project)	✅	✅
Update stage/upload photo	❌	✅ (own project)	❌	✅
Raise complaint	✅	—	View/respond	View/respond
Onboard new project	❌	❌	Request	✅
Cross-project analytics	❌	❌	✅	✅

6.4 Notifications

- SMS + push on stage change (SMS fallback critical — many residents may not have smartphones)

- Alert to govt dashboard if a project has no update for 30+ days

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7. Non-Functional Requirements

- **Accessibility:** Must work on low-end Android devices, low bandwidth (2G/3G tolerant); SMS/IVR fallback for status checks
- **Language:** Marathi, Hindi, English at minimum
- **Data integrity:** Photo uploads must be tamper-evident (timestamp + geotag); periodic physical inspection by SRA officer for spot-verification
- **Privacy:** Resident personal data (Aadhaar, family details) encrypted at rest; role-based access strictly enforced
- **Auditability:** Every stage update is immutable and logged with who/when/from-where
- **Uptime:** Not mission-critical infra, but should target 99% during business hours

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8. Success Metrics (Pilot, 6 months)

- ⩾70% of registered families check the app/SMS status at least monthly
- 100% of pilot project milestones updated within 7 days of actual completion
- Reduction in ward-office in-person status inquiries by 40%
- Government dashboard used to generate at least 1 formal status report for assembly reference

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9. Rollout Plan

Phase	Scope	Duration
Phase 0	Design + data model finalization, contractor onboarding for 1 project	4 weeks
Phase 1 (Pilot)	2–3 projects, 500–1000 families, 1 ward	3 months
Phase 2	Expand to full BMC ward / SRA zone	6 months
Phase 3	State-wide SRA/MHADA integration, dashboard for Housing Dept	12+ months

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10. Open Questions / Risks

- **Data source of truth:** Will the app pull eligible-family lists directly from existing SRA records, or require fresh registration? (Recommend: integrate with SRA's existing dweller database where possible to avoid duplicate data entry.)
- **Contractor incentive:** What compels a contractor to update honestly? (Recommend: tie milestone-update compliance to payment release approvals.)
- **Ownership post-pilot:** Who hosts/maintains the app long-term — ward office, SRA, or a state IT body? This affects tech stack and hosting decisions later.
- **Verification of self-reported progress:** Photos alone can be gamed; may need periodic third-party/SRA inspector confirmation for high-value milestones (foundation, structure completion).

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11. Recommended Tech Stack (for engineering handoff)

- **Mobile app:** React Native (single codebase for resident + contractor apps, works on low-end Android)
- **Backend:** Node.js/Express or Django REST Framework, PostgreSQL (relational hierarchy fits Project→Building→Unit well)
- **File storage:** S3-compatible object storage for photos, with metadata (geotag, timestamp) stored alongside
- **Notifications:** Firebase Cloud Messaging (push) + SMS gateway (e.g., MSG91/Twilio) for fallback
- **Admin/Govt dashboard:** React web app with map view (Mapbox/Google Maps) for project locations
- **Hosting:** Cloud-agnostic (AWS/Azure) but should ultimately target NIC/state government cloud (MeghRaj) for long-term public-sector deployment compliance